FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

FIITJEE

JEE(Advanced)-2016

ALL INDIA INTEGRATED TEST SERIES

ANSWERS, HINTS & SOLUTIONS
HALF COURSE TEST–VIII
(Paper - 1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**B**	**A**	**D**
2.	**A**	**B**	**A**
3.	**C**	**D**	**B**
4.	**A**	**A**	**A**
5.	**A**	**C**	**B**
6.	**B**	**D**	**C**
7.	**A, B, C**	**B, C, D**	**A, C**
8.	**A, B, D**	**A, B, C, D**	**A, B, C, D**
9.	**A, B, C, D**	**A, B, D**	**A, B, C, D**
10.	**D**	**C**	**C**
11.	**A**	**A**	**B**
12.	**A**	**D**	**D**
13.	**A**	**D**	**C**
14.	**C**	**A**	**C**
1.	**(A) → (s), (B) → (p) (C) → (p), (D) → (q, r)**	**(A) → (r); (B) → (q); (C) → (p); (D) → (s, t)**	**(A) → (s) (B) → (p) (C) → (r,t) (D) → (p,q,r,t)**
2.	**(A) → (r), (B) → (q) (C) → (q), (D) → (s)**	**(A) → (p, s); (B) → (p, q, r, t); (C) → (p, q, t); (D) → (p)**	**(A) → (p, s) (B) → (p, r) (C) → (p) (D) → (p)**
1.	**6**	**8**	**1**
2.	**3**	**3**	**2**
3.	**6**	**4**	**1**
4.	**5**	**8**	**4**
5.	**5**	**5**	**5**

Physics

PART – I

SECTION – A

1. $R = \frac{N}{t} = a\frac{P}{T}\sqrt{T} = a\frac{P}{\sqrt{T}}$

Now we can see that when temperature increases 4 times and pressure increases 8 times, the rate of outflow of the gas increases 4 times.

2. The vertical component of acceleration of block 1 and block 2 are

$a_1 = g\sin^2 60°$, $a_2 = g\sin^2 30°$

Since vertical displacement for both blocks is 1 m, the block with larger acceleration will reach the base of wedge first. Hence block of mass m_1 shall reach base of wedge first.

3. Let length of semi major axis is a and eccentricity is e.

$$-\frac{GMm}{R_e} + \frac{1}{2}m\left(\sqrt{\frac{GM_e}{R_e}}\right)^2 = -\frac{GM_em}{2a} \Rightarrow a = R_e$$

As the separation between ends of semi minor axis and focus is equal to length of semi major axis we can say that launching and landing sites are ends of semi minor axis.

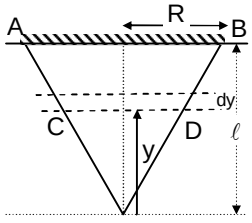
$\Rightarrow R_e\cos 45° = ae \Rightarrow e = 1/\sqrt{2}$

Then maximum separation will be $a + ae = R_e\frac{\left(\sqrt{2}+1\right)}{\sqrt{2}}$

4. $T = 2\pi\sqrt{\frac{\frac{mr^2}{4} + mx^2}{mgx}}$, put $\frac{dT}{dx} = 0$ we get $x = \frac{r}{2}$

5. Weight of OCD part = $m(y/L)^3 g$

Radius of CD = $(y/\ell)R$



$$\text{Stress at CD} = \frac{m\left(\frac{y}{\ell}\right)^3 g}{\pi\left\{\left(\frac{y}{\ell}\right)R\right\}^2}$$

$$\text{Elastic potential energy per unit volume at CD} = \frac{1}{2}\frac{1}{Y}\left[\frac{m\left(\frac{y}{\ell}\right)^3 g}{\pi\left\{\left(\frac{y}{\ell}\right)R\right\}^2}\right]^2$$

$$\text{Total elastic potential energy} = \int_0^{\ell}\frac{1}{2}\frac{1}{Y}\left[\frac{m\left(\frac{y}{\ell}\right)^3 g}{\pi\left\{\left(\frac{y}{\ell}\right)R\right\}^2}\right]^2 \pi\left\{\left(\frac{y}{\ell}\right)R\right\}^2 dy = \frac{m^2g^2\ell}{10\pi R^2 Y}$$

6. $\frac{1}{2}mv^2 = \int_{\frac{3\lambda}{4}}^{L+\frac{3\lambda}{4}} F_0 \cos\left(\frac{2\pi x}{\lambda}\right)dx = \frac{F_0\lambda}{2\pi}\int_{\frac{3\lambda}{4}}^{L+\frac{3\lambda}{4}} \cos y\, dy = \frac{F_0\lambda}{2\pi}\left[\sin\left(\frac{2\pi L}{\lambda}+\frac{3\pi}{2}\right)-\sin\frac{3\pi}{2}\right] = \frac{F_0\lambda}{2\pi}\left(1-\cos\frac{2\pi L}{\lambda}\right)$

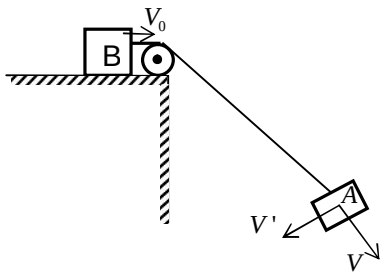
$v = \sqrt{\frac{F_0\lambda}{m\pi}\left(1-\cos\frac{2\pi L}{\lambda}\right)}$

7. Pressure force is balanced by spring force. Work done & change in internal energy can be calculated by energy conservation.

8. Apply conservation of momentum and equation of coefficient of restitution.

9. Node should be at clamped ends and antinode should be at free end.

11. Let velocity of block B at that moment be V. component of velocity of A along string at that instant is also V and component perpendicular to string ray V', then applying conservation of angular momentum about pulley $m\,V'\,2l = mV_0 l$



$\Rightarrow V' = V_0/2$

Applying conservation of energy

Initial KE = Final KE

$\frac{1}{2}mV^2 + \frac{1}{2}\left(V^2 + V'^2\right) = \frac{1}{2}mV_0^2$

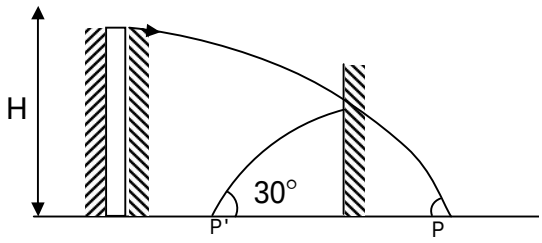
Solving, $V = V_0\sqrt{\frac{3}{8}}$.

After B strikes pulley component velocities along string becomes zero and component perpendicular to string remains unchanged

$V' = \frac{V_0}{2}$.

SECTION – C

1. AP is the motion of projectile in absence of wall, which is half of a normal projectile with range R = 2 OP and angle of projection $\theta = 30°$ and height H



$BP' = OP' = \frac{1}{2}OB = BP$

$\Rightarrow OP = \frac{3}{2}OB$

$\Rightarrow R = 12\sqrt{3}m$

$H = \frac{u^2\sin^2\theta}{2g}$

$R = \frac{u^2\sin 2\theta}{g}$

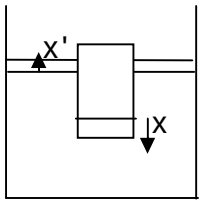
$\Rightarrow H = \frac{1}{2}\left(\frac{\sin^2\theta}{\sin 2\theta}\right)R = 6$

2. Particle will cross the square if it crosses the point of maximum potential between A and C–which is C itself.

$$\frac{1}{2}mv_0^2 = K + \frac{1}{4\pi\varepsilon_0}\frac{4Q^2q}{(l/\sqrt{2})}$$

$$K > 0 \Rightarrow V_0^2 = 8\sqrt{2}\left(\frac{1}{4\pi\varepsilon_0}\right)\left(\frac{Qq}{ml}\right)$$

3. Cylinder can perform SHM only till it is partially submersed. When cylinder goes down x in ground from liquid level comes up by x' (say)



$(4a - a)x' = xa \Rightarrow x' = x/3$

So the centre of the cylinder goes down by (w.r.t. the liquid surface).

$$(x + x') = \frac{4}{3}x \le \frac{l}{10} \Rightarrow x \le \frac{3l}{40} = 6cm\,.$$

4. Constant velocity means zero acceleration, both block work under dynamic equilibrium.

For B,

$T = f = \mu mg$

$$\Rightarrow \quad \frac{T}{\mu} = 5$$

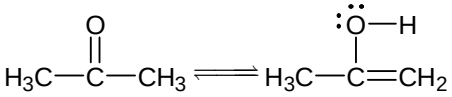
5. $$v^2 = \frac{T}{\mu} = \frac{3Mg}{\mu} + gx$$

$$a = v\frac{dv}{dx} = \frac{g}{2} = 5m/s^2\,.$$

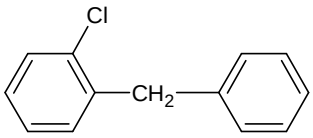
Chemistry

PART – II

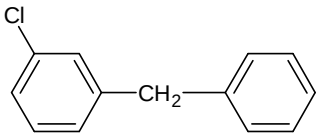
SECTION – A

2. 

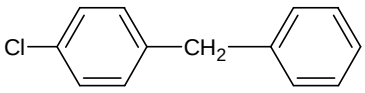
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3. 

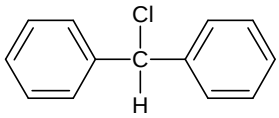
(I)



(I)



(III)



(IV)

4. $H_2SO_4 \rightleftharpoons H^+ + HSO_4^-$

$H^+ + OH - NO_2 \longrightarrow H_2O + \underset{\substack{\text{(nitronium ion)}\\ \text{electrophile}}}{\overset{+}{N}O_2}$

5. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

t_0	1	0
t_{eq}	$1-\alpha$	2α

$\sum n = 1-\alpha+2\alpha = 1+\alpha$

$\Delta n_g = 2 - 1 = 1$

$K_p = \frac{(2\alpha)^2}{(1-\alpha)} \times \left(\frac{P}{1+\alpha}\right)^1 \Rightarrow K_p = \frac{4\alpha^2}{1-\alpha^2}.P$

$\alpha = \left[\frac{K_p/P}{4+K_p/P}\right]^{1/2}$

6. $A_2X_3 \rightleftharpoons 2A^{3+} + 3X^{2-}$

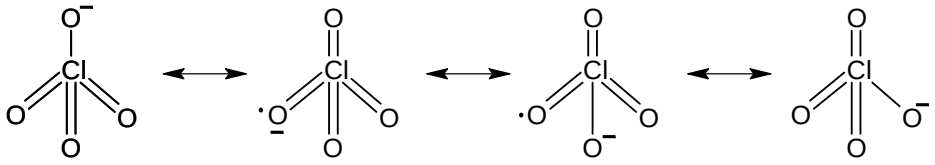
2s 3s

$K_{sp} = [A^{3+}]^2[X^{2-}]^3 = [2s]^2[3s]^3 = 108\ s^5$

8. (a) $r \propto \frac{n^2}{Z}$, (b) P.E. $\propto \frac{-Z^2}{n^2}$, (c) K.E. $\propto \frac{Z^2}{n^2}$, (d) $v \propto \frac{Z}{n}$

12. CH_4 has no π - bond.

13.



$$\text{B.O.} = \frac{\text{total bond on Cl in any resonating structure}}{\text{total number of resonating structure}} = \frac{7}{4} = 1.75$$

SECTION – C

1.

$$Mg(OH)_2 \rightleftharpoons Mg^{2+} + 2OH^-$$
$$x \quad 2x+3y$$
$$Al(OH)_3 \rightleftharpoons Al^{3+} + 3OH^-$$
$$y \quad 3y+2x$$

$$K_{sp} = 4\times10^{-12} = x(3y+2x)^2 \text{ and } 10^{-33} = y(3y+2x)^3$$

$$\frac{x}{y} = 8\times10^{17}\times10^{-17} = 8$$

Mathematics

PART – III

SECTION – A

1. $D_1 + D_2 + D_3 = a^2 + c^2 + d^2 \geq 0$

2. To be perfect square, both the given equations have equal roots

$\therefore \quad \frac{1}{1} = -\frac{(\lambda - 2)}{\lambda} = \frac{\lambda^2}{2\lambda - 1}$

$\therefore \quad \lambda = 1$

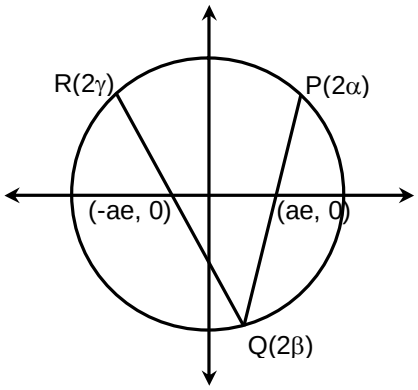
3. $a = \cos\theta$, $b = \sin\theta$

$\therefore \quad h = \sqrt{\cos^2\theta - \sin^2\theta}$, $k = \frac{\sin^2\theta}{\cos\theta}$

$\therefore$ locus, is

$2y^2(x^2 + 1) = (x^2 - 1)^2$

4. Equation of PQ : $\frac{x}{a}\cos(\alpha+\beta) + \frac{y}{b}\sin(\alpha+\beta) = \cos(\alpha-\beta)$



$e = \frac{\cos(\alpha-\beta)}{\cos(\alpha+\beta)}$

$e = -\frac{\cos(\beta-\gamma)}{\cos(\beta+\gamma)}$

$\therefore \quad \frac{\cos(\alpha-\beta)}{\cos(\alpha+\beta)} = -\frac{\cos(\beta-\gamma)}{\cos(\beta+\gamma)}$

Applying, component and dividend

$\tan\alpha \cdot \tan\gamma = \cot^2\beta$.

6. Let, P(h, 0) be point on x-axis apply

$T = S_1$, and passes through the point $(a(\lambda - 1), a(\lambda + 1))$

We get

$\lambda^2 - 12\lambda - 4 > 0$.

7. $\tan\alpha \cdot \tan\beta = -1$

$\Rightarrow \quad \alpha - \beta = \frac{\pi}{2}$.

8. Let the slope of u = 0 be m then the slope of v = 0 is $\frac{9m}{2}$

$\therefore \quad \frac{7}{9} = \left|\frac{m - \frac{9m}{2}}{1 + m\cdot\frac{9m}{2}}\right| \Rightarrow m = \pm\frac{1}{3}, \pm\frac{2}{3}$.

10. Clearly, $D_1 = D_2$

$\therefore \quad -\frac{D_1}{4} = -\frac{1}{4} \Rightarrow D_1 = D_2 = 1$

least value of $g(x) = -\frac{1}{4}$

11. $-\frac{b_2}{2} = \frac{7}{2} \Rightarrow b_2 = -7.$

12-14. $\alpha = \frac{t+4}{2}$

$\Rightarrow \quad t = 2\alpha - 4$

$\beta = \frac{5+\sin t}{2}$

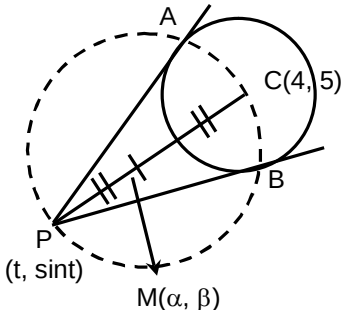
$\Rightarrow \quad = \frac{5\sin(2\alpha-4)}{2}$

$\therefore \quad y = \frac{5+\sin(2x-4)}{2}$

$\therefore$ Range of $f(x) = [2, 3]$

Period of $f(x) = \pi$

Range of $f^{-1}(x) = \left[-\frac{\pi}{4}+2, \frac{\pi}{4}+2\right]$



SECTION – B

1. (A) If, $x < 2$

$\therefore \quad x = -2, 4$

no solution

If, $-2 \le x < 3$

$\therefore \quad x = 2, -2$

two solution

If, $x \ge 3$

$\therefore \quad x = 4$

one solution

$\therefore \quad N = 3$

(B) $x = 1$, satisfies

$\therefore \quad \lambda = -2$

$\therefore \quad [\lambda - 1] = -3$

(C) $-3x^2 - 3x - 3 < x^2 - \lambda x - 2 < 2x^2 + 2x + 2$

solving, we get

$\lambda \in (-1, 2)$

$\therefore \quad [\lambda] = -1, 0, 1$

(D) $f(x) = x^3 - 6x^2 + 9x + \lambda$

$f(1) = \lambda + 4$

$f(3) = \lambda$

$f'(x) < 0$ in $(1, 3)$

$\therefore \quad f(1)\, f(3) < 0$

$\therefore\ \lambda(\lambda + 4) < 0$

$\therefore\ [\lambda + 1] = -3, -2, -1, 0.$

2. (A) $\begin{vmatrix} 1 & -2 & -6 \\ 3 & 1 & -4 \\ \lambda & 4 & \lambda^2 \end{vmatrix} = 0 \Rightarrow \lambda = 2, -4$

(B) $\begin{vmatrix} \lambda+1 & 1 & 1 \\ 2\lambda+1 & 3 & 1 \\ 2\lambda+2 & 2\lambda & 1 \end{vmatrix} = 0 \Rightarrow \lambda = 2, -1/2.$

(C) Point of intersection of $x - y + 1 = 0$
and $3x + y - 5 = 0$, is $(1, 2)$
$\therefore\ 1 + 2 - 1 - \lambda = 0 \Rightarrow \lambda = 2.$

(D) Mid-point of $(1, -2)$ and $(3, 4)$ will satisfy $y - x - 1 + \lambda = 0 \Rightarrow \lambda = 2.$

SECTION – C

1. $\Delta = \frac{1}{2}\begin{vmatrix} b & c & 1 \\ c & a & 1 \\ a & b & 1 \end{vmatrix} = \frac{1}{2}\ (ab + ca + ab - a^2 - b^2 - c^2)$

$\Delta' = (a + b + c)^2\,\Delta \Rightarrow \frac{\Delta'}{256\Delta} = 1$

2. $2a + 3b - 5 = 0$
$\therefore$ line is at greatest distance from centre
$\therefore\ \left(\frac{4-3}{3-2}\right)\left(-\frac{a}{b}\right) = -1 \Rightarrow a = b$
$\therefore\ |a + b| = 2$

3. Let m be the slope of line
$\therefore\ m \times \left(\frac{2t + 2 - t}{(t-1) - (2t+1)}\right) = -1$
$\therefore\ m = 1.$

4. Equation of the circle is
$(x - r - 1)^2 + y^2 = r^2$
putting $y^2 = 4x$ and $D = 0$
$\therefore\ r = 4$

5. Shortest distance along common normal, slope of common normal = $0, \pm 2$
$\therefore$ feet of normals will be $(4, 4)$ and $(5, 2)$ or $(4, -4)$ and $(5, -2)$
$\therefore\ d = \sqrt{1+4} = \sqrt{5} \Rightarrow d^2 = 5$